

Objectives

- Define SPA and its benefits
- Define React and identify its working
- Identify the differences between SPA and MPA
- Explain Pros & Cons of Single-Page Application
- Explain about React
- Define virtual DOM
- Explain Features of React

In this hands-on lab, you will learn how to:

- Set up a react environment
- Use create-react-app

Prerequisites

The following is required to complete this hands-on lab:

- Node.js
- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **30 minutes**.

Create a new React Application with the name “myfirstreact”, Run the application to print “welcome to the first session of React” as heading of that page.

1. To create a new React app, Install Nodejs and Npm from the following link:

<https://nodejs.org/en/download/>

2. Install Create-react-app by running the following command in the command prompt:

```
C:>npm install -g create-react-app
```

3. To create a React Application with the name of “myfirstreact”, type the following command:

```
C:>npx create-react-app myfirstreact
```

4. Once the App is created, navigate into the folder of myfirstreact by typing the following command:

```
C:>cd myfirstreact
```

5. Open the folder of myfirstreact in Visual Studio Code
6. Open the App.js file in Src Folder of myfirstreact
7. Remove the current content of "App.js"
8. Replace it with the following:

```
function App() {  
  return (  
    <h1> Welcome the first session of React </h1>  
  );  
}
```

9. Run the following command to execute the React application:

```
C:\myfirstreact>npm start
```

10. Open a new browser window and type "localhost:3000" in the address bar



```
PS C:\Users\anana\Downloads\Digital Nurture\Week 5\React\1. ReactJS-HOL> npm install -g create-react-app
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if
you want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerfu
l.
npm warn deprecated fstream-ignore@1.0.5: This package is no longer supported.
npm warn deprecated uid-number@0.0.6: This package is no longer supported.
npm warn deprecated rimraf@2.7.1: Rimraf versions prior to v4 are no longer supported
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security vulnerabi
lities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorb
itant rates) by contacting i@izs.me
npm warn deprecated fstream@1.0.12: This package is no longer supported.
npm warn deprecated tar@2.2.2: Old versions of tar are not supported, and contain widely publicized security vulnerabi
lities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorb
itant rates) by contacting i@izs.me

added 64 packages in 6s

4 packages are looking for funding
  run 'npm fund' for details
npm notice
npm notice New minor version of npm available! 11.9.0 -> 11.18.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.18.0
npm notice To update run: npm install -g npm@11.18.0
npm notice
PS C:\Users\anana\Downloads\Digital Nurture\Week 5\React\1. ReactJS-HOL> npx create-react-app myfirstreact
create-react-app is deprecated.
```

```
Windows PowerShell
This error message will only be shown once per install.

Creating a new React app in C:\Users\anana\Downloads\Digital Nurture\Week 5\React\1. ReactJS-HOL\myfirstreact.

Installing packages. This might take a couple of minutes.
Installing react, react-dom, and react-scripts with cra-template...

added 1301 packages in 1m

272 packages are looking for funding
  run 'npm fund' for details

Installing template dependencies using npm...

added 17 packages, removed 2 packages, and changed 1 package in 9s

271 packages are looking for funding
  run 'npm fund' for details
Removing template package using npm...

removed 1 package, and audited 1316 packages in 9s

271 packages are looking for funding
  run 'npm fund' for details

28 vulnerabilities (9 low, 6 moderate, 13 high)

To address issues that do not require attention, run:
```

```
Windows PowerShell
Happy hacking!
PS C:\Users\anana\Downloads\Digital Nurture\Week 5\React\1. ReactJS-HOL> cd myfirstreact
PS C:\Users\anana\Downloads\Digital Nurture\Week 5\React\1. ReactJS-HOL\myfirstreact> npm start

> myfirstreact@0.1.0 start
> react-scripts start

(node:20552) [DEP0176] DeprecationWarning: fs.F_OK is deprecated, use fs.constants.F_OK instead
(Use 'node --trace-deprecation ...' to show where the warning was created)
(node:20552) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
(node:20552) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBeforeSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
Starting the development server...
Compiled with warnings.

[eslint]
src\App.js
  Line 1:8:  'logo' is defined but never used  no-unused-vars

Search for the keywords to learn more about each warning.
To ignore, add // eslint-disable-next-line to the line before.

WARNING in [eslint]
src\App.js
  Line 1:8:  'logo' is defined but never used  no-unused-vars

webpack compiled with 1 warning
```

```
JS App.js
src > JS App.js > App
1  import logo from './logo.svg';
2  import './App.css';
3
4  function App() {
5    return (
6      <h1> Welcome the first session of React</h1>
7    );
8  }
9
10 export default App;
11
```

